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SATHYABAMA INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

CONTINUOUS ASSESSMENT EXAM - 2- SET-B

Programme: B.E –ECE/EEE/MECH/MECHATRONICS/AERO. Max. Marks: 50
Course Code: S11BLH11 Duration: 2 Hours
Course Name: Problem Solving Techniques Using C Date : 21.11.2024
Batch: 2023 - 2027 Sem 3

Part-A Answer Any Five (5 ×10 =50)

Q. No.	Questions	CO
1.	Write a C program to calculate the final price a customer pays at a retail store based on their membership type and total purchase amount. The store offers three membership levels: Platinum, Gold, and Silver, with base discounts of 20%, 15%, and 10%, respectively. Use a switch-case statement to determine the base discount by membership type. If the total purchase amount is less than $100, no additional discount is applied. For purchases between $100 and $200, an additional 5% discount is added, and for purchases above $200, an additional 10% discount is added. Use if-else statements to apply the additional discount based on the total purchase amount, then display the final price. If an invalid membership type is entered, display an error message.	CO3
2.	John is creating a feedback application for his restaurant. The program should prompt five customers to rate their dining experience on a scale of 1 to 5, where 5 is excellent and 1 is poor. The program should use an array to store each customer's rating. After collecting all ratings, the program should calculate and display the average rating. Write a C program that uses a `for` loop to collect ratings and calculate the average.	CO4
3.	Create a C program that simulates a fuel tracking system for a vehicle. The program should start with an initial fuel level of 50 liters and prompt the user to enter the number of fuel transactions they want to log. For each transaction, display a menu with options: Refuel, Record Consumption, or Check Fuel Level. Use a switch-case statement to handle each option.	CO3

	Refuelling should add a positive amount to the fuel level and display the refuel cost based on a fuel price of 100 currency units per liter. Recording consumption should subtract a specified amount from the fuel level, provided it does not exceed the current level, and checking the fuel level should display the current amount of fuel. After each transaction, display the estimated mileage that can be covered based on a mileage rate of 15 km per liter. At the end, display the final fuel level and estimated mileage.	
4.	Write a C program to calculate the sum of the main diagonal elements of a 3x3 matrix. Prompt the user to enter the values for each element in the matrix. Calculate the sum of the elements along the main diagonal, then display both the original matrix and the sum of its diagonal elements.	CO4
5.	Write a C program that allows the user to perform a single operation on an integer array: **Insert, Delete, Search,** or **Sort.** First, ask the user for the size of the array and its elements. Based on the chosen option, perform the selected operation on the array and display the result.	CO3
6.	**Answer all the questions with explanations, each carrying 2 marks:** **a.** What is the output of the following code snippet? *int a = 3;* *int b = 7;* *if (++a > 3 || b-- < 5) {* *printf("%d %d\n", a, b);* *} else {* *printf("%d %d\n", a, b);* *}* *A) 4 7* *B) 4 6* *C) 3 6* *D) 3 7* **b.** What will be the output of the following code? *int x = 5;* *while (++x < 10);* *printf("%d\n", x);* *A) 9* *B) 10* *C) 11* *D) 5*	CO4

c. What will be the result of the following code?

```
int i = 10;
do {
    printf("%d ", i);
    i++;
} while (i < 15);
```

A) Infinite loop B) 10 11 12 13 14 C) 10 11 12
D) No output

d. What will be the result of the following code?

```
for (int i = 1; i <= 3; i++) {
    for (int j = 1; j <= 3; j++) {
        if (i + j == 4) continue;
        printf("i=%d, j=%d\n", i, j);
    }
}
```

A) Skips when i+j=4 B) Skips when i+j=3
C) Skips when i+j=2 D) Skips all i+j pairs

e. What will be the result of the following code?

```
int x = 1, y = 0;
if (x || y) {
    x += 2;
} else {
    y += 2;
}
printf("%d %d\n", x, y);
```

A) 1 0 B) 3 0 C) 1 2 D) 3 2

7.	Write a C program that prompts the user to enter a positive integer as a starting number and a lower limit. The program should display a countdown from the starting number to the lower limit, showing only the odd numbers within this range. If the user enters a negative number or if the starting number is less than the lower limit, display an error message and prompt the user to re-enter both values using goto to jump back to the input section. The program should continue prompting for new values and displaying results until the user enters -1 as the starting	CO3

	number to exit.	
8.	**Answer all the questions with explanations, each carrying 2 marks:** **f.** What is the output of the following code snippet? *int arr[] = {1, 2, 3, 4, 5};* *for (int i = 0; i < 5; i++) {* *arr[i] += arr[i + 1];* *}* *printf("%d\n", arr[0]);* *A) 12* *B) 18* *C) 15* *D) 9* **g.** What will be the output of the following code? *int a[] = {0, 1, 2, 3, 4};* *for (int i = 4; i >= 0; i -= 2) {* *printf("%d ", a[i]);* *}* *A) 4 3 2* *B)4 20* *C) 4 1* *D) 4 3* **h.** What will be the result of the following code? *int arr[3][3] = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};* *printf("%d", arr[2][1]);* A) 5 B) 6 C) 8 D) 7 **i.** What will be the result of the following code? *int arr[5] = {1, 2, 3, 4, 5};* *int *ptr = arr + 1;* *printf("%d\n", *(ptr + 2) + *(ptr - 1));* A) 5 B) 6 C) 8 D) 7 **j.** What will be the result of the following code? *int x = 1, y = 0;* *if (x || y) {* *x += 2;* *} else {* *y += 2;* *}* *printf("%d %d\n", x, y);* *A) 1 0* *B) 3 0* *C) 1 2* *D) 3 2*	CO4